IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 29 July 2024, accepted 20 September 2024, date of publication 16 October 2024, date of current version 26 November 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3481410

RESEARCH ARTICLE

DCE-YOLOv8: Lightweight and Accurate Object Detection for Drone Vision

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This work was supported in part by Korea Institute for Advancement of Technology (KIAT) Grant funded by Korea Government (MOTIE) [Development of Regional Innovation Cluster (RIC)] under Grant P0024175, and in part by the Regional Innovation Strategy (RIS) through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (MOE) under Grant 2021RIS-003.

ABSTRACT Object detection using drones is a sophisticated technology that employs a camera mounted on a drone in conjunction with a computer vision algorithm to pinpoint the precise location of an object and ascertain its type. Drones are capable of rapidly scanning extensive areas, thereby facilitating efficient data collection and analysis. This capability can yield critical information and bolster swift response efforts. The utilization of object detection technology in drones offers numerous advantages. Nevertheless, despite the benefit of drones' ability to swiftly scan wide areas, several challenges persist, including image resolution, the detection of small-sized objects, overlapping objects, and concentrated distributions. In this paper, we introduce DCE-YOLOv8, an advanced model based on YOLOv8. DCE-YOLOv8 is engineered to address the low detection rate of small objects in drone imagery. To effectively detect small objects, it is imperative to either enhance the resolution of drone images or efficiently extract the features of small objects. Additionally, the efficient integration of these extracted features is crucial. The ERB(Efficient Residual Bottleneck) and DCE(Divided Context Extraction) modules are incorporated into the Backbone, with the ERB module reducing the number of parameters to render the model more lightweight. The DCE module focuses on extracting features pertinent to small objects. Subsequently, the rate of missed detections is mitigated by comprehensively merging the shallow and deep features extracted from the neck part. The proposed method is trained using the VisDrone, demonstrating superior detection performance compared to other state-of-the-art methods. When comparing the proposed method with the YOLOv8 small version using the VisDrone dataset, the mean Average Precision value improved by approximately 43%, increasing from 22.8mAP to 32.7mAP, while the number of parameters decreased by about 57%, from 11,166,560 to 4,822,382. The Average Inference Time per Image has been optimized to 11.4 ms, which is relatively slower than YOLOv8's 5.9 ms, yet it still maintains a robust frame rate of 87.71 FPS, emphasizing its potential for real-time detection applications. Furthermore, the proposed method underwent additional experiments using the TT100K and AFO datasets. Compared to YOLOv8 small, it demonstrates superior performance while maintaining a comparable average inference time. This paper holds significant value in balancing object detection rates and real-time operational speed, serving as a guiding reference for in-depth research in related fields.

INDEX TERMS Object detection, divided context extraction, lightweight, drone vision, YOLOv8.

I. INTRODUCTION

Object detection constitutes a sophisticated technique aimed at identifying and pinpointing specific objects within images or videos. This task represents a cornerstone in the domain

The associate editor coordinating the review of this manuscript and approving it for publication was Szidonia Lefkovits.

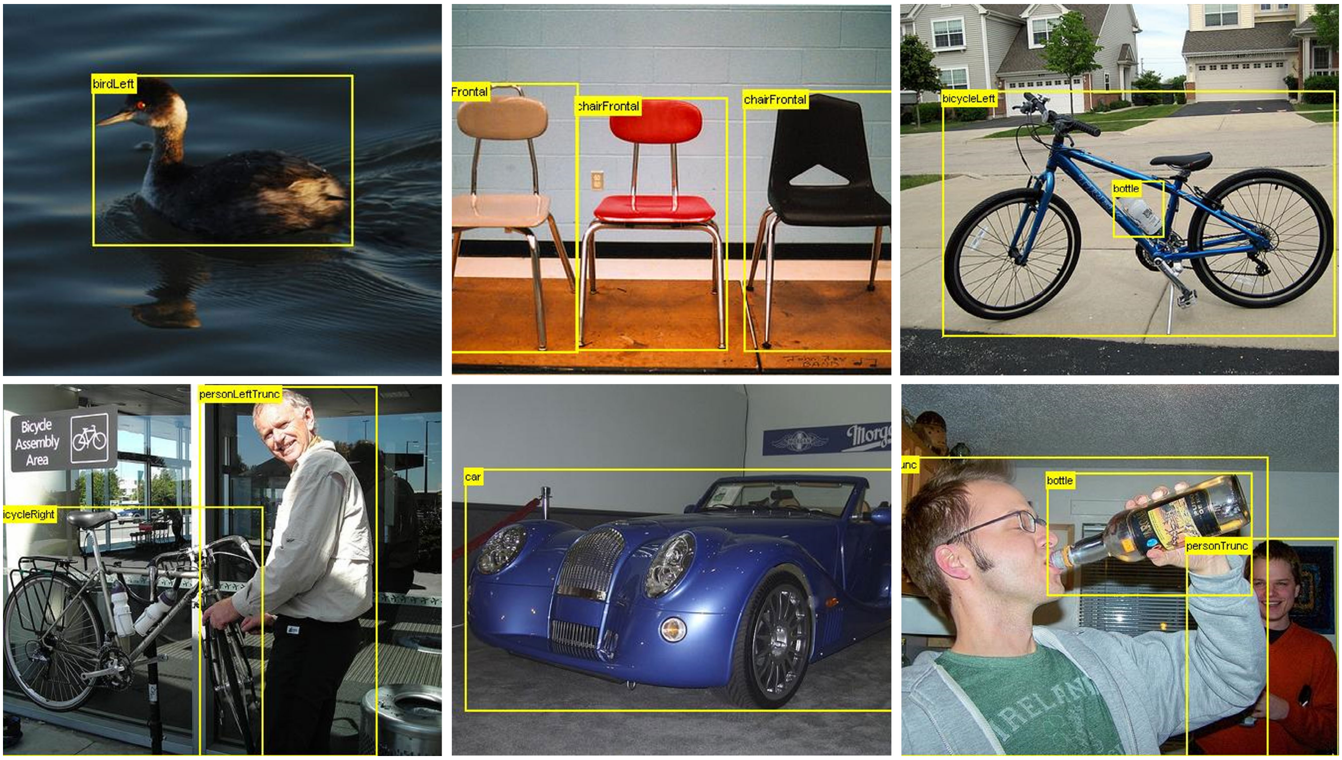


FIGURE 1. **Sample Images of PASCAL VOC dataset captured by people on the ground.**

of computer vision and finds applicability across a diverse array of fields. Contemporary deep neural network models employed for object detection are predominantly developed and validated using meticulously curated image datasets. The PASCAL-VOC2007 [1], VOC2012 [2], and MS-COCO [3] datasets, exemplified in Figure 1, encompass image data captured from a human perspective and exhibit the following traits: (1) The number of objects per image is relatively limited. (2) Objects are spaced apart, with minimal overlapping occurrences. (3) Objects occupy a substantial portion relative to the overall image size. (4) The majority of images are captured from a head-on perspective. (5) The backgrounds are relatively static. (6) Lighting conditions are predominantly optimal.

A drone, or unmanned aerial vehicle (UAV), is an aircraft piloted via computer or remote control. The advent of drone technology is spearheading transformative innovations across a myriad of industries. These unmanned aircraft, capable of autonomous flight without a ground-based pilot, offer numerous benefits including high mobility, precise detection, data collection, and seamless communication. Recently, drones have found applications in diverse sectors such as reconstruction, aerial photography, logistics, agriculture, and rescue operations. Outfitted with high-resolution cameras and vacuum stabilization systems (Gimbals), drones are extensively employed in aerial photography, enabling the capture of unique and dynamic visuals for film production, advertisements, news coverage, and sports events. Drone photography also serves as an effective tool for promoting real estate and land properties. In the logistics sector, drones are gaining traction as a rapid and efficient delivery solution. They prove especially advantageous in delivering goods to hard-to-reach areas or during emergencies, facilitating the swift transportation of medical supplies or food to disaster-stricken regions. Within agriculture, drones play a pivotal role in monitoring crop growth, detecting diseases and pests, and forecasting crop yields. Furthermore, the application of fertilizers and pesticides via drones is emerging as a cost-effective and efficient method. Drones are indispensable in search and rescue missions in inaccessible disaster zones or hazardous locations. Equipped with infrared cameras and thermal sensors, drones can support rescue operations even in nocturnal or low-visibility conditions. Real-time transmission of situational data to rescue teams enables rapid and informed responses.

Drone Vision denotes a cutting-edge technology that captures and analyzes imagery through cameras and sensors affixed to drones. Integral to the drone vision system is computer vision, a technology that processes and comprehends this image data. Computer vision, a domain leveraging machine learning, pattern recognition, and image processing, enables computers to interpret and understand visual information. This encompasses tasks such as object detection, classification, tracking, and segmentation. Once image data of the surrounding environment is acquired via cameras or other sensors mounted on the drone, computer

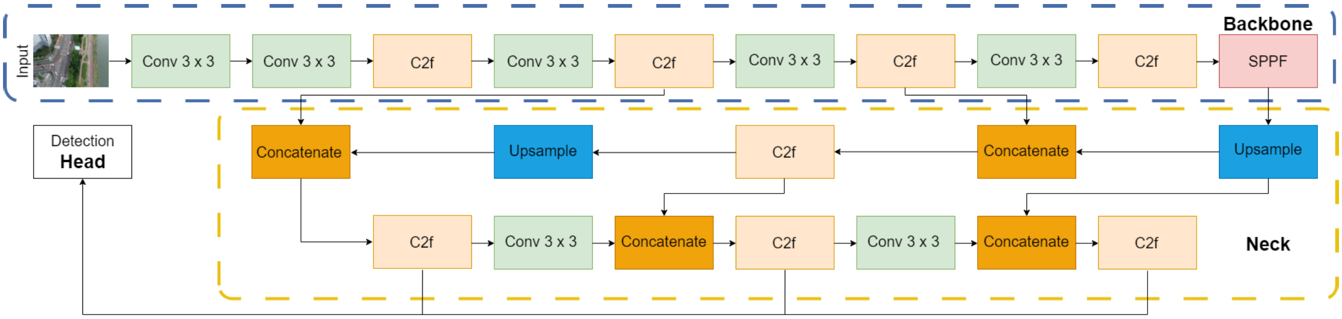


FIGURE 2. YOLOv8 Architecture consists of three main components, Backbone, Neck, and Head.

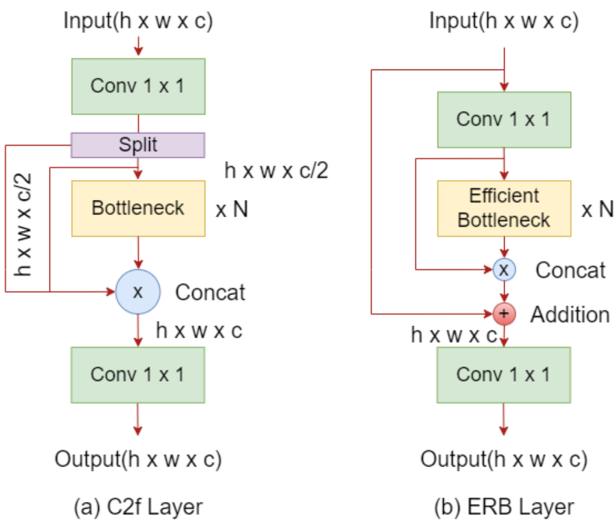


FIGURE 3. Comparison of C2f layer and ERB layer. (a) C2f layer of YOLOv8 network (b) Efficient Residual Bottleneck layer of DCE-YOLOv8 for making network lightweight and fast.

vision technology processes this information. Drone vision finds applications across a multitude of fields. For instance, drone vision technology facilitates real-time environmental recognition and the autonomous planning of flight paths based on this data. This capability proves invaluable in rescue operations and disaster response scenarios. Additionally, it is employed in agriculture to monitor crop health, detect pests and diseases, and predict harvest yields. By analyzing crop conditions, computer vision algorithms can significantly enhance agricultural productivity. Furthermore, drone vision systems are adept at surveilling extensive areas, detecting intruders or abnormal activities in real-time, thereby bolstering the efficiency of security systems. The synergy between drone vision and computer vision is driving innovation across various industries, and its potential is poised to expand into even more application domains in the future.

The majority of objects captured by drones are diminutive in size. Consequently, the real-time detection of these small objects is of paramount importance. YOLOv8, a preeminent real-time object detection algorithm, typically demonstrates superior performance; however, its accuracy diminishes when detecting small objects. Furthermore, it encounters challenges in effectively detecting objects across a wide range of scales. This issue is particularly pronounced when both large and small objects are present simultaneously, making it difficult for the model to maintain consistent performance across all scales.

In this paper, we introduce an enhanced neural network model predicated on the YOLOv8 architecture, adept at meeting data processing demands with respect to the detection accuracy of small and tiny objects. The principal contributions of this paper are delineated as follows:

1) We introduce an advanced iteration of YOLOv8, termed DCE-YOLOv8. The proposed model outperforms the performance of the original YOLOv8 network by about 43%. This refined model boasts a reduced parameter count, exhibits superior detection rates for diminutive objects, and maintains the capability for real-time operation.
2) Drawing upon the C2f layer employed in YOLOv8 and the C3 layer of YOLOv5, we propose a novel layer termed the Efficient Residual Bottleneck (ERB). This layer not only renders the network more lightweight but also facilitates the regulation of interactions among extracted image features.
3) An efficient extraction module termed Divided Context Extraction (DCE), grounded in partial convolution [4] is proposed to distinguish between target object and background rapidly. This module adeptly integrates shallow and deep features, thereby significantly enhancing the detection rate of small-sized objects.
4) In order to augment feature extraction with an emphasis on small and tiny objects, down-sampling is replaced with up-sampling, and the model's weight is diminished by excising the B5 block.

II. RELATED WORK

A. CNN

A Convolutional Neural Network (CNN) is a specialized type of artificial neural network tailored for image and video processing. CNNs are architected to proficiently learn the spatial hierarchies of images. Through convolutional and pooling layers, CNNs generate feature maps that encapsulate the salient features of images. Leveraging these feature maps, CNNs execute tasks such as object classification, detection, estimation, and segmentation.

B. YOLO

YOLO [5] is a real-time object detection model rooted in CNN technology, capable of concurrently detecting and classifying multiple objects within an image. "You Only Look Once" (YOLOv8), the latest iteration in the YOLO series illustrated in Figure 2, boasts several enhanced features. YOLOv8 primarily comprises three components: Backbone, Neck, and Head. The Backbone is responsible for extracting features from the input image. The backbone consists of several convolution layers and C2f layers. The backbone of YOLOv5 consists of C3, and in YOLOv8 it was improved to C2f. The C3 module in YOLOv5 stands for the CSP (Cross Stage Partial) Bottleneck with 3 convolutions. and the C2 represents CSP Bottleneck with 2 convolutions. It is a building block used in the YOLOv8 architecture. The C2f module is a faster implementation of the C2 module. And the combination of Convolution and C2f is called the Backbone Block(B2∼B5).

Developed as an advancement of YOLOv5, YOLOv8 incorporates a total of five significant modifications. Firstly, the C3 module has been replaced with C2f, which offers a comparable performance with fewer parameters. Secondly, the $Conv_{6\times6}$ in the Backbone has been substituted with $Conv_{3\times3}$, effectively halving the parameter count due to the reduced kernel size. Thirdly, layers No.10 and No.14 from the YOLOv5 architecture have been removed. Fourthly, the $Conv_{1\times1}$ bottleneck has been replaced with $Conv_{3\times3}$. Lastly, the decoupled head structure has been adopted, eliminating the objectness branch. Objectness is a loss technique that determines an object's confidence score, and the decoupled head separates the regression and classification tasks.

Additional distinctions include the anchor-free detection methodology. Unlike YOLOv5, which sets several anticipated Bbox initial values and resizes anchor values to determine the actual object's size, YOLOv8 directly predicts the object's center without initial values. This approach accelerates Non-Maximum Suppression (NMS) by reducing the number of Bbox predictions, thereby enhancing efficiency.

1) C2F LAYER

The C2f layer in YOLOv8 is an integral component of the network's backbone. Utilized within the Modified CSPDarknet53 [6] (backbone network), the C2f layer adeptly merges feature maps of varying sizes contingent on the network's depth. It serves as an efficient mechanism to modulate the dimensions of feature maps within the network. To this end, the C2f layer alternates between 1×1 and 3×3 convolutions to generate feature maps of diverse sizes. This architecture incorporates gradient shunt connections, thereby enhancing the information flow within the feature extraction network while preserving its lightweight nature.

The operational dynamics of the C2f layer are as follows:

Initially, it generates feature maps of assorted sizes via 1×1 convolution of the input feature map. Subsequently, the feature map is bisected and processed with 3×3 convolution. This step captures features at multiple scales while retaining the spatial information of the features. The process of alternating between 1×1 and 3×3 convolutions is repeated several times, enabling the integration of feature maps of different sizes and facilitating a comprehensive understanding of objects' varying dimensions and characteristics.

The C2f layer's application within the Modified CSP-Darknet53 backbone network proves highly effective for object detection tasks in YOLOv8. By flexibly managing feature maps of different sizes, the C2f layer enhances object detection accuracy. Consequently, YOLOv8 demonstrates the capability to efficiently detect objects of various sizes.

2) PANET

PANet [7] (Path Aggregation Network) is an instance segmentation model derived from Mask R-CNN. This model, which incorporates a neck structure demonstrating superior performance, is also employed in YOLOv5. The neck component is crucial for enhancing object detection accuracy by amalgamating and harmonizing feature maps of varying sizes.

PANet comprises two distinct pathways: bottom-up and top-down. The Bottom-Up Pathway is designed to convey low-level feature information to higher levels, aiming to preserve low-level details by minimizing information loss during transmission. Low-level features are generally advantageous for detecting small objects, but they are also essential for detecting large objects, particularly in responding strongly to edges or small instances. Conversely, detecting small objects also requires high-level features that encapsulate the image's contextual information. This pathway extracts feature maps at multiple scales from a given input image, where smaller-sized feature maps capture broader contextual information and larger-sized feature maps retain more detailed object information.

The Top-Down Pathway initiates with a small-sized feature map and incrementally increases the feature map size. This process enhances the detailed information of objects by leveraging high-level features derived from small-sized feature maps. PANet adeptly integrates feature maps of diverse sizes to bolster object detection accuracy. Furthermore, it is engineered to efficiently process both low-level and high-level feature information, thereby facilitating a comprehensive understanding of objects' sizes and attributes.

III. PROPOSED METHOD

YOLOv8 stands as the quintessential network for object detection. The architecture of YOLOv8 is composed of three primary components: Backbone, Neck, and Head. The Backbone is tasked with extracting the essential features required for object detection from images. These extracted features are then transmitted through the Neck to the Head,

TABLE 1. Parameters of backbone in YOLOv8 network, why remove B5 block.

C	From	n	Parameters	Module	Arguments
0	-1	1	928	Conv	[3,32,3,2]
1	-1	1	18,560	Conv	[32,64,3,2]
2	-1	1	29,056	C2f	[64,64,1,True]
3	-1	1	73,984	Conv	[64,128,3,2]
4	-1	2	197,632	C2f	[128,128,2,True]
5	-1	1	295,424	Conv	[128,256,3,2]
6	-1	2	788,480	C2f	[256,256,2,True]
7	-1	1	1,180,672	Conv	[256,512,3,2]
8	-1	1	1,838,080	C2f	[512,512,1,True]
9	-1	1	656,896	SPPF	[512,512,5]

TABLE 2. Parameters of backbone in DCE-YOLOv8 network.

C	From	n	Parameters	Module	Arguments
0	-1	1	1,856	Conv	[3,64,3,2]
1	-1	1	20,736	DCE	[64,4,'split_cat']
2	0	1	73,984	Conv	[64,128,3,2]
3	-1	1	82,944	DCE	[128,4,'split_cat']
4	2	1	98,944	ERB	[128,128,1]
5	-1	1	18,048	SCDown	[128,128,3,2]
6	-1	2	172,928	ERB	[128,128,2]
7	-1	1	36,096	SCDown	[128,256,3,2]
8	-1	2	689,920	ERB	[256,256,2]
9	-1	1	164,608	SPPF	[256,256,5]

where the actual object detection takes place. The YOLOv8 network is adept at identifying small, medium, and large-sized objects, and its architecture has been refined to enhance the detection of small and tiny objects. This paper proposes four methodologies for improvement: (1) Efficient Residual Bottleneck, (2) Divided Context Extraction, (3) removal of the B5 block, and (4) replacing down-sampling with up-sampling.

A. THE BACKBONE

In this session, we will delve into the Backbone, one of the three core components of YOLOv8. The Backbone is responsible for extracting image features and transmitting them to the Head via the Neck. This paper primarily aims to enhance the object detection rate in drone imagery by emphasizing the detection of small and tiny objects. The extraction of high-quality image features is crucial for object detection, particularly at a low level. High-quality image features encapsulate the prominent visual characteristics of an object, encompassing various attributes such as shape, color, and texture. Effective extraction of these features enables precise prediction of an object's bounding box location and size, thereby facilitating accurate identification of the object's position and classification.

B. REMOVAL OF B5 BLOCK

YOLOv8 employs three feature map sizes—Large, Medium, and Small—for object detection. These feature maps are derived through a series of continuous convolutional down-sampling operations, as illustrated in Figure 2, which depicts the structure of YOLOv8's feature extraction network. As indicated in Table 1, the B5 block (positions No.7 and No.8) comprises a significant number of parameters within the respective layer. The total number of parameters in the YOLOv8 backbone is 5,079,712, and the number of parameters in the B5 block is 3,018,752, accounting for approximately 60% of the total number of parameters in the backbone. To streamline the model, the B5 block has been removed. The revised structure of the feature extraction network for DCE-YOLOv8 can be observed in Table 2. The total number of parameters of the DCE-YOLOv8 backbone with the B5 block removed is 1,360,064, which is about 27% of the YOLOv8 backbone.

C. EFFICIENT RESIDUAL BOTTLENECK

The Efficient Residual Bottleneck is a layer engineered to enhance the existing C3 layer of YOLOv5 and the C2f layer of YOLOv8. The C3 and C2f layers are CSP Bottlenecks with 3 and 2 convolutions, respectively, comprising bottleneck and several convolutional layers. To enable real-time operation of the object detection network on drones, it is essential to reduce the network's parameters. The details of this module are illustrated through the following equation:

$$ERB = Conv_{1\times1}(F(Bottleneck_{Eff}(Conv_{1\times1}(X_1)), X_2)+X_1) \tag{1}$$

Here X_1 represents the input feature, and X_2 is the input for the efficient bottleneck. We achieve this by adjusting the number of convolutional layers and incorporating an addition operation to optimize the layer's efficiency. By minimizing the number of convolutional layers, we reduce both the number of parameters and computational load, thereby

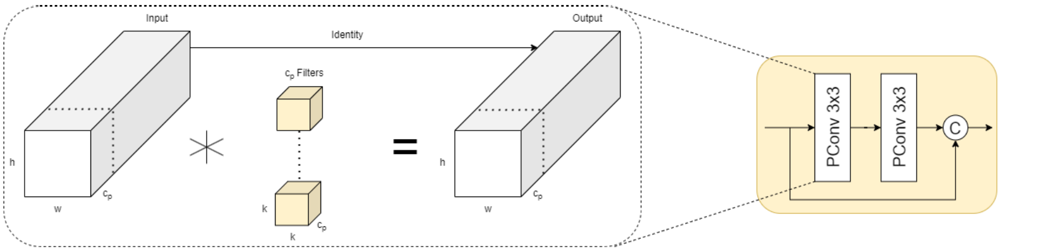


FIGURE 4. **Divided Context Extraction Module for extracting low-level feature information focusing on the tiny and small size object.**

enhancing the model's speed. Furthermore, the addition operation facilitates the modulation of interactions between extracted image features. The residual technique is employed to preserve the quality of feature maps, culminating in superior performance in final predictions. To prevent gradient degradation and avoid saturation during the training process, each convolution operation sequentially employs SiLU activations and batch normalization.

D. DIVIDED CONTEXT EXTRACTION

The Divided Context Extraction module represents an enhanced convolutional layer designed to extract image features at a low level. Instead of utilizing Regular Conv, this module employs Partial Conv (PConv), which demands less computation and memory access compared to regular convolution operations.

Convolution operations traditionally entail extensive filter layers, resulting in a substantial number of parameters and computational bottlenecks that impede and slow down the inference process. In this study, we introduce the Divided Context Extraction module, which aims to reduce the reliance on extensive channel map operations and extractions. The input map (X) is divided so that three-fourths of the features are processed through sequential convolution operations, while the remaining one-fourth is directed towards the feature fusion process. The details of this module are illustrated through the following equation:

$$\begin{aligned} DCE &= F(Conv_{3\times3}(Conv_{3\times3}(X_1), i) \\ &= F(Conv_{3\times3}((X_2), i) \end{aligned} \tag{2}$$

Here, X_1, X_2 and i represent the first and second divided features and identity, respectively. The majority of features undergo the extraction process, selectively separating the object context. These features are prioritized for extraction due to their rich coarse information content. The context separation process involves a sequential 3×3 convolution operation $Conv_{3\times3}$, which optimizes the filter. Batch normalization and SiLU activation are associated with each convolution operation to enhance the extraction performance. Feature fusion F at different levels is applied at the end of the module to combine information from various frequencies, thereby enriching the feature diversity while retaining the input features as residual information.

E. REPLACE DOWN-SAMPLING TO UP-SAMPLING

As previously mentioned, YOLOv8 utilizes three feature map sizes—Large, Medium, and Small—for object detection. In this paper, we introduce a feature map one size larger to enhance the detection of small and tiny objects. Given that most objects captured via drones are relatively small, we opted to employ feature maps of tiny, small, and medium sizes instead of the conventional small, medium, and large sizes. Tiny-sized objects are detected by integrating the feature map extracted through the aforementioned Divided Context Extraction with the up-sampled feature map.

F. LOSS FUNCTION

The loss function of YOLOv8 is instrumental in training the model by quantifying the discrepancy between the model's predictions and the ground truth in object detection. To enhance both the location and class prediction accuracy, YOLOv8 employs multiple loss functions. The loss function in YOLOv8 primarily comprises two branches: a classification branch utilizing Binary Cross Entropy (BCE) loss, and a regression branch that integrates two distinct losses—Distribution Focal Loss (DFL) [8] and Complete Intersection over Union (CIoU) [9].

DFL is designed to address the class imbalance issue prevalent in object detection tasks and is also leveraged in YOLOv8 to refine bounding box (BBox) regression, particularly for objects with indistinct or ambiguous boundaries. Instead of predicting a fixed BBox, DFL estimates the probability distribution of BBox coordinates, thereby mitigating uncertainty in their location. Conversely, CIoU loss not only considers the overlap between the predicted and ground truth boxes but also accounts for the aspect ratio disparity between them. The synergy of these two loss functions significantly enhances the performance of the regression task, especially when dealing with small objects.

In contrast to YOLOv5, YOLOv8 omits the objectness loss, which previously assessed the model's confidence in the existence of an object within a BBox by comparing the predicted probability with the ground truth. In YOLOv8, objectness confidence is embedded within the IoU recognition classification score, eliminating the necessity for a separate loss function. The classification branch facilitates the network's ability to accurately predict the class of detected objects.

In summary, YOLOv8 employs BCE loss for classification, while BBox regression is driven by a combination of DFL and CIoU loss. The final loss function is a weighted sum of these three individual losses, with the weights modulating the relative importance or contribution of each loss in the overall combined loss, which is optimized during training.

$$L_{total} = w_1 BCE_{loss} + w_2 CIoU_{loss} + w_3 DFL_{loss} \quad (3)$$

IV. IMPLEMENTATION DETAILS

This session outlines experiments conducted using the DCE-YOLOv8 network, which integrates ERB and DCE, on the VisDrone, TT100K, and AFO datasets. The experiments are carried out in a computing environment based on Linux Ubuntu 20.04, utilizing AMD EPYC 7302 CPU and an NVIDIA Tesla A30 24G graphics card. Python version 3.10 is employed alongside the PyTorch deep learning framework. To optimize computational resource usage and adjust video memory capacity, the batch size is set to 2 for hyperparameter configuration, while other hyperparameters remained at their default settings. The input image size is configured to the standard 640 $\times$ 640, as per the network's requirements. To reflect the efficiency of the proposed network and facilitate comparisons with existing models, the default settings of YOLOv8 are maintained. The optimizer is configured to SGD, with a learning rate of 0.01, momentum set to 0.937, warmup epochs of 3, and a total of 200 epochs.

V. EVALUATION AND EXPERIMENTS

Four comprehensive evaluation methodologies were devised to assess the performance of DCE-YOLOv8. The first methodology involves comparing the performance and parameter count with both the original YOLO series and its enhanced variants. The results of this evaluation are presented in Table 3. The second methodology focuses on comparing the performance, inference time, FPS, and parameter count of DCE-YOLOv8 with the YOLO series and other modified networks. To ensure a fair comparison, inference times were measured using the same GPU across all modified networks. The results of this comparison can be found in Table 4, demonstrating that all models are capable of real-time operation. The third methodology illustrates the parameters, performance, inference time, and FPS of DCE-YOLOv8 according to its sequential improvements. These results are detailed in Table 5, which confirms the enhancement of detection rates while maintaining a balance with real-time performance. The fourth methodology compares the performance of DCE-YOLOv8 across different datasets. Table 6 compares the results of the proposed method with those of other YOLO series on the TT100K and AFO datasets. In Table 6, DCE-YOLOv8 exhibits superior performance compared to other YOLO series while demonstrating a lower parameter count, confirming its capability for real-time operation.

The proposed network achieves commendable results when applied to the detection of small and tiny objects. Object detection in drone tasks necessitates that the vision system precisely localizes and classifies objects on a diminutive scale. This presents a significant challenge due to the low resolution of the objects, which provides information from only a limited set of pixels. Consequently, the extractable information is constrained. Our framework addresses this by preserving small object features through the utilization of a dual pyramid system, which maintains the quality of information pertaining to small objects. As a result, our model is capable of detecting small objects captured by drones. Objects such as persons, bicycles, and motorcycles, measuring less than 30 $\times$ 30 pixels, are categorized as low resolution. Figure 6 illustrates the prediction visualizations of the YOLO series and our proposed network on the VisDrone dataset. And the prediction visualizations of YOLOv8 small alongside the proposed network on the TT100K and AFO datasets are displayed in Figure 7.

A. DATASET

This section provides an overview of the datasets utilized for training and evaluation, highlighting the relevance and characteristics of each dataset. We trained and evaluated the proposed network using the VisDrone, TT100K, and AFO datasets. As a result, the network demonstrates superior performance compared to the comparative networks across all three datasets, while also maintaining real-time operational capability.

1) VISDRONE

The VisDrone [10] dataset represents a comprehensive, large-scale collection designed to facilitate object detection, a critical domain within computer vision research, using images captured by aerial drones. This dataset is versatile, supporting various applications such as object detection, object tracking, and visual tracking, and encompasses images taken across diverse environments and conditions, including urban areas, agricultural fields, and highways.

With 10 distinct object classes meticulously annotated, the dataset features a variety of categories such as people, vehicles, bicycles, buildings, trucks, buses, tricycles, motorcycles, boats, traffic lights, and traffic signs. These objects appear in a myriad of sizes and positions within the video frames, thereby enabling robust research for object detection and tracking across different scenarios.

The dataset includes images captured at various resolutions and viewpoints (ranging from high altitude to low altitude), and under numerous visual conditions, such as the presence of small and large objects, densely packed objects, and sparsely distributed objects.

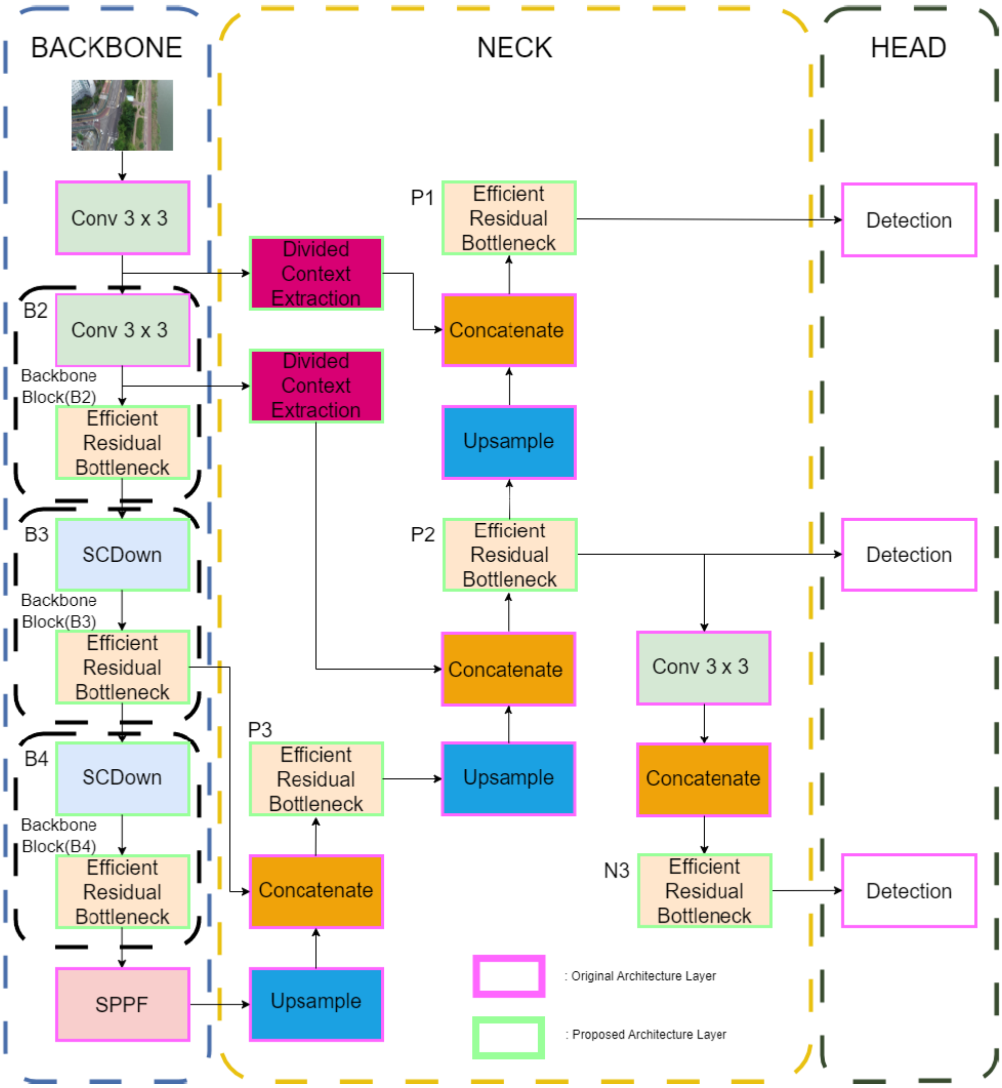


FIGURE 5. The proposed method architecture called DCE-YOLOv8.

By providing a standardized benchmark, the VisDrone dataset significantly aids computer vision and machine learning researchers in developing and evaluating diverse algorithms. Its high-resolution images and variety of objects across different environments present realistic and challenging problems, thereby driving technological advancements and performance improvements. Consequently, the VisDrone dataset has made substantial contributions to computer vision research and is esteemed as a pivotal resource with broad applications.

2) TT100K

The TT100K dataset represents a comprehensive, large-scale collection meticulously crafted to advance the field of object detection within urban environments. It encompasses a diverse array of images captured in various settings, including streets, highways, and public spaces, rendering it an invaluable resource for researchers and practitioners in the domain of computer vision. Comprising 100,000 images annotated with 40 distinct object classes, the dataset includes categories such as cars, buses, bicycles, pedestrians, traffic signs, and more. The extensive annotations not only classify objects but also provide precise bounding boxes, facilitating accurate localization within the frames. The variety of objects, characterized by differing sizes, orientations, and occlusions, presents a formidable challenge for object detection algorithms.

TT100K features images captured from multiple angles and perspectives, reflecting real-world conditions such as varying lighting, weather, and traffic densities. This diversity enhances the dataset's applicability for training and evaluating algorithms in realistic scenarios, where objects may appear in complex arrangements.

By establishing a standardized benchmark, the TT100K dataset significantly bolsters researchers' efforts in developing and assessing various object detection models. Its extensive annotations and high-quality imagery promote the exploration of advanced techniques aimed at improving detection accuracy and robustness. Consequently, the TT100K dataset has made significant contributions to the advancement of computer vision research, serving as a pivotal resource with far-reaching applications in intelligent transportation systems and autonomous driving technologies.

3) AERIAL DATASET OF FLOATING OBJECTS

The Aerial dataset of Floating Objects is a meticulously curated collection designed to advance the study of object detection in aerial imagery, specifically focusing on objects situated in water bodies and similar environments. This dataset serves as an invaluable resource for researchers aiming to develop and evaluate algorithms capable of accurately identifying and tracking floating objects, which present unique challenges due to their varied appearances and dynamic contexts. Comprising a diverse array of images, the dataset includes annotations for a range of floating objects such as boats, debris, and wildlife, among others. The detailed annotations provide precise bounding boxes and object classifications, facilitating accurate localization and recognition within the imagery. The floating objects exhibit a spectrum of sizes, shapes, and orientations, reflecting the complexities inherent in real-world scenarios.

Captured from various altitudes and angles, the images in this dataset represent different environmental conditions, including fluctuating lighting, diverse water surface patterns, and the presence of reflections. This diversity enhances the dataset's applicability for training and testing detection algorithms under realistic conditions, where environmental factors may partially obscure or influence floating objects.

By establishing a standardized benchmark, the Aerial Dataset of Floating Objects significantly supports researchers in propelling the field of computer vision and machine learning forward. Its high-resolution imagery and comprehensive annotations present realistic and challenging problems, thus driving innovation and advancements in detection technologies. Consequently, this dataset has made substantial contributions to the understanding of object detection in aerial contexts. It has established itself as a pivotal resource with extensive applications in maritime surveillance, environmental monitoring, and related fields.

B. ABLATION STUDIES

This session delineates the variations in parameters and object detection rates as each enhancement is incrementally applied. The experimental results are detailed in Table 5. The YOLOv8 small version serves as the baseline, with a parameter count of 11,166,560. The mAP.5:.95 is 23.3 and the mAP 0.5 is 39.2.

1) PERFORMANCE IMPROVEMENT

The B5 block is a significant component within the YOLOv8 architecture, comprising around 60% of the total parameters in the backbone network, with a total parameter count of approximately 3,018,752. By removing this block, the overall parameter count is significantly reduced by about 44%, bringing it down to 7,390,846. This reduction is not just about decreasing complexity; it also enhances performance. Specifically, the mean Average Precision (mAP) at IoU 0.5 improves by about 21% to reach 47.6, while the mAP.5:.95 increases by roughly 25% to 29.2. This change demonstrates that eliminating the B5 block can both simplify the model and substantially improve its object detection capabilities.

The Divided Context Extraction (DCE) module introduces an advanced method for extracting image features while optimizing computational efficiency. Unlike conventional convolution operations, DCE uses Partial Convolution (PConv), which requires less computation and memory access, thereby accelerating the inference process. The input feature map is divided so that a majority of features (three-fourths) pass through sequential convolution operations, while the remainder goes directly to a feature fusion process. This approach leverages the strengths of various convolutional layers to extract features effectively, enhancing the detection of both small and large objects. As a result, the DCE module elevates model performance, increasing mAP.5:.95 by 6% to 32.6, even with an increased parameter count of 7,226,040.

2) LIGHTWEIGHT NETWORK

The Efficient Residual Bottleneck (ERB) is designed to bolster the model's efficiency while minimizing the number of parameters and computational load. By streamlining the number of convolutional layers and incorporating an effective addition operation, the ERB not only contributes to a reduction in the overall parameter count to about 4,524,536 but also enhances the model's performance. This leads to a 5% increase in mAP.5:.95, now achieving a score of 30.7, and a 4% increase in mAP 0.5, climbing to 49.7. The ERB's architecture also addresses potential issues like gradient degradation, ensuring that the model remains robust even during rigorous training.

SCDown is a novel strategy aimed at further reducing parameters while maintaining similar performance levels. After implementing SCDown, the parameter count drops to approximately 4,822,382, marking a reduction of about 44%. The performance metrics, particularly mAP.5:.95 and mAP 0.5, remain stable at around 32.7 and 52.7, respectively. This stability underscores SCDown's effectiveness in preserving performance while decreasing the model's complexity. By replacing traditional down-sampling methods with SCDown, the architecture showcases an impressive balance between efficiency and detection accuracy, making it particularly suitable for applications requiring real-time inference.

C. EVALUATION OF THE PERFORMANCE ON DATASET

To assess the performance of various models in object detection, we select 11 networks for comparison: YOLOv5, YOLOv6, YOLOv7, YOLOv8, ACAM-YOLO, MS-YOLOv7,

TABLE 3. Experimental results on VisDrone2019 dataset.

Method	mAP0.5	mAP.5:.95	Params (M)
YOLOv5-n [11]	33.1	19.2	2.5
YOLOv5-s [11]	39.3	23.4	9.1
YOLOv5-l [11]	41.4	24.4	46.5
YOLOv6-s [12]	35.1	20.3	18.5
YOLOv7-l [13]	47.1	26.4	71.4
YOLOv8-s [14]	39.5	23.5	11.1
YOLOv8-l [14]	43.7	26.9	76.7
ACAM-YOLO [15]	49.5	29.6	15.9
MS-YOLOv7 [16]	53.1	31.3	79.7
EdgeYOLO [17]	44.8	26.4	40.5
SF-YOLOv5 [18]	34.3	18.2	2.24
EL-YOLOv5	31.6	18.4	7.6
Drone-YOLO(Nano) [19]	38.1	22.7	3.0
Drone-YOLO(Small) [19]	42.8	25.6	5.4
Drone-YOLO(Large) [19]	51.3	31.9	76.2
UAV-YOLOv8s [20]	47.0	29.2	10.3
SODCNN [21]	54.1	32.0	32.1
DCE-YOLOv8(ours)	**52.7**	**32.7**	**4.8**

TABLE 4. Experimental results on VisDrone2019 dataset based on the GPU devices.

Method	mAP.5:.95	mAP0.5	Average Inference Time per Image	FPS	Params(M)	GPUs
YOLOv5-s	23.4	39.3	**5.5 ms**	**181.81**	9.1	A30
YOLOv8-s	23.5	39.5	5.9 ms	169.49	11.1	A30
SF-YOLOv5	17.0	31.9	6.8 ms	147.05	**2.1**	A30
EL-YOLOv5(small)	18.4	31.6	9.5 ms	105.37	7.6	A30
Drone-YOLO(nano)	22.7	38.1	12.3 ms	81.10	3.1	A30
DCE-YOLOv8(ours)	**32.7**	**52.7**	11.4 ms	87.71	4.8	A30
UAV-YOLOv8	23.6	40.0	**19.5 ms**	**51.2**	10.3	RTX 3060
DCE-YOLOv8(ours)	**32.7**	**52.7**	20.0 ms	50.0	**4.8**	RTX 3060

TABLE 5. Detection outcomes following the implementation of various enhancement strategies. The bolded data in the table signify the optimal results.

BaseLine	wo B5 block	ERB	DCE	SCDown	Params	mAP.5:.95	mAP 0.5	Inference Time (ms)	FPS
YOLOv8 small					11,166,560	23.3	39.2	**5.9**	**169.49**
	✓				7,390,846	29.2	47.6	9.1	109.89
	✓	✓			**4,524,536**	30.7	49.7	8.7	114.94
	✓	✓	✓		7,226,040	32.6	52.4	18.6	53.76
	✓	✓	✓	✓	4,822,382	**32.7**	**52.7**	11.4	87.71

TABLE 6. Detection results on other datasets with NVIDIA A30.

Datasets	Method	Size	Params (M)	mAP.5:.95	mAP 0.5	Average Inference Time per Image	FPS
TT100K	YOLOv5-s	640	7.1	60.7	79.7	11.81 ms	84.7
	UCN-YOLOv5-s [22]	640	7.4	65.1	85.6	22.32 ms	44.8
	YOLOv6-l	640	59.6	69.6	88.3	28.57 ms	35.0
	YOLOv8-s	640	11.2	74.3	**86.7**	7 ms	142.86
	DCE-YOLOv8(ours)	640	**4.8**	**75.9**	86.6	**6.9 ms**	**144.93**
AFO	YOLOv5-s	640	2.1	60.0	94.4	**3.6 ms**	**277.77**
	YOLOv8-s	640	11.1	61.7	95.1	5.5 ms	181.81
	DCE-YOLOv8(ours)	640	**2.0**	**62.2**	**95.5**	5.6 ms	178.57

EdgeYOLO, SF-YOLOv5, Drone-YOLO, UAV-YOLOv8, SODCNN. It is juxtaposed with DCE-YOLOv8 for a thorough evaluation. Uniformity of evaluation is maintained as all models are trained and tested on the VisDrone dataset. Performance metrics encompassed mAP0.5, mAP.5:.95, and Params(M). The experimental results of object detection are encapsulated in Table 3.

The analysis of the results reveals that the DCE-YOLOv8 model when compared with the aforementioned models, outperforms in terms of mAP.5:.95 with a score of 32.7. Although the parameters of DCE-YOLOv8 are greater than those of YOLOv5-n, SF-YOLOv5, and Drone-YOLO(nano), they are significantly smaller than those of the other models.

Given the prevalence of small-sized objects, high-resolution images or techniques that adeptly extract object features are essential. We introduce a method utilizing the Divided Context Extraction module to enhance the accuracy of small object feature extraction. This approach yields an improved object detection performance, achieving a 32.7 mAP, which marks approximately a 43% enhancement over the original YOLOv8 small network. Moreover, the number of parameters is reduced by about 57%, totaling 4,822,382.

Additionally, experiments were conducted on the TT100K and AFO datasets, with the results presented in Table 6.

For the TT100K dataset, the YOLO series (versions 5, 6, and 8) and UCN-YOLOv5 were selected as comparative networks. As shown in Table 6, DCE-YOLOv8 has the fewest parameters at 4,855,654 compared to other networks. It also achieves the highest mAP at 75.9, with an inference time of 6.9 ms and an FPS of 144.93, demonstrating robust real-time capabilities.

For the AFO dataset, a comparison was made with YOLOv8. DCE-YOLOv8 exhibits approximately 18% of the parameters of YOLOv8s, totaling 2,080,421 while achieving an mAP of 62.2. The inference time is 5.6 ms, which is slightly slower than YOLOv8s by 1 ms; however, it maintains sufficient real-time performance with an FPS of 178.57.

D. EVALUATION OF MODEL SPEED

In this section, we present a comparative analysis of the proposed DCE-YOLOv8 model against the YOLOv5 small and YOLOv8 small models, focusing on speed and accuracy metrics. The metrics evaluated include mAP0.5, mAP.5:.95, Average inference time per image, FPS, and Parameters(M), as detailed in Table 4

DCE-YOLOv8 is a lightweight and accurate network that strikes a well-balanced compromise between real-time performance and object detection accuracy. While it exhibits a longer inference time compared to the YOLOv5 small and YOLOv8 small models, it sufficiently meets the real-time object detection requirements in terms of Frames Per Second (FPS). Typically, real-time processing necessitates achieving at least 30 FPS, and the DCE-YOLOv8 model achieves approximately 87 FPS with an average inference time of 11.4 ms. This clearly demonstrates the model's suitability for real-time applications.

Furthermore, the DCE-YOLOv8 model showcases exceptional performance in terms of accuracy, with mAP0.5 and mAP.5:.95 values of 52.7 and 32.7, respectively. This impressive accuracy underscores the model's ability to maintain high performance while effectively handling real-time processing.

In terms of model complexity, the DCE-YOLOv8 model is highly efficient, with only 4.8M parameters. This is significantly lower than the 9.1M parameters of YOLOv5 small and the 11.1M of YOLOv8 small. This efficiency highlights the lightweight and optimized architecture of the DCE-YOLOv8 model.

In summary, the DCE-YOLOv8 model offers a well-rounded performance profile, excelling in speed, accuracy, and parameter efficiency, making it an optimal choice for real-time object detection applications.

E. DISCUSSION OF WEAKNESS OF DETECTION

This section discusses the advantages and weaknesses of the proposed detector. A general challenge in object detection is achieving tight localization of the object and precise label prediction. Drone vision often has difficulty recognizing tiny objects due to limited information in small pixels, which requires careful feature extraction to ensure important information is not lost. Additionally, the prediction layer requires an effective learning method to handle a variety of similar object features. While the DCE-YOLOv8 model demonstrates superior performance in terms of accuracy and efficiency, it is important to discuss potential weaknesses observed during detection tasks. Figure 6 illustrates the detection results of three models: YOLOv5 small, YOLOv8 small, and the proposed DCE-YOLOv8. It shows that the proposed detector better locates objects than the competitors, even for tiny objects that are difficult for the human eye to see. This demonstrates the strength of our model in effectively and precisely determining the coordinates and size of objects from drone footage. Moreover, the generated models require fewer trained weights, which speeds up the training time and weight calculation, resulting in a shorter training phase. The DCE-YOLOv8 model consistently outperforms the YOLOv5 small and YOLOv8 small models, as reflected in the higher mAP0.5 and mAP.5:.95 values. However, despite its advantages, certain limitations are evident. Here are the detailed limitations of our detector:

1) FALSE POSITIVES AND NEGATIVES

Even though the DCE-YOLOv8 model shows higher overall accuracy, it is not immune to false positives and false negatives. In some complex scenes with overlapping objects or cluttered backgrounds, the model may incorrectly identify objects (false positives) or miss detecting them altogether (false negatives). This can be seen in Figure 6, where certain objects are either misclassified or not detected by any of the models, albeit less frequently in DCE-YOLOv8. This is because the objects are similar to the background, and our detectors are confused in distinguishing them.

2) SMALL OBJECT DETECTION

Detecting small objects remains a challenge for most object detection models, including DCE-YOLOv8. While the model performs relatively better in identifying smaller objects compared to YOLOv5 small and YOLOv8 small, there are still instances where small objects are either missed or inaccurately detected. This limitation can be observed in Figure 6, where smaller objects in the scene are sometimes not detected. The small input size of objects in our network results in information being completely lost in the high-level features. This feature loss forces the detection head to ignore the object.

(a) YOLOv5s (b) YOLOv8s (c) DCE-YOLOv8

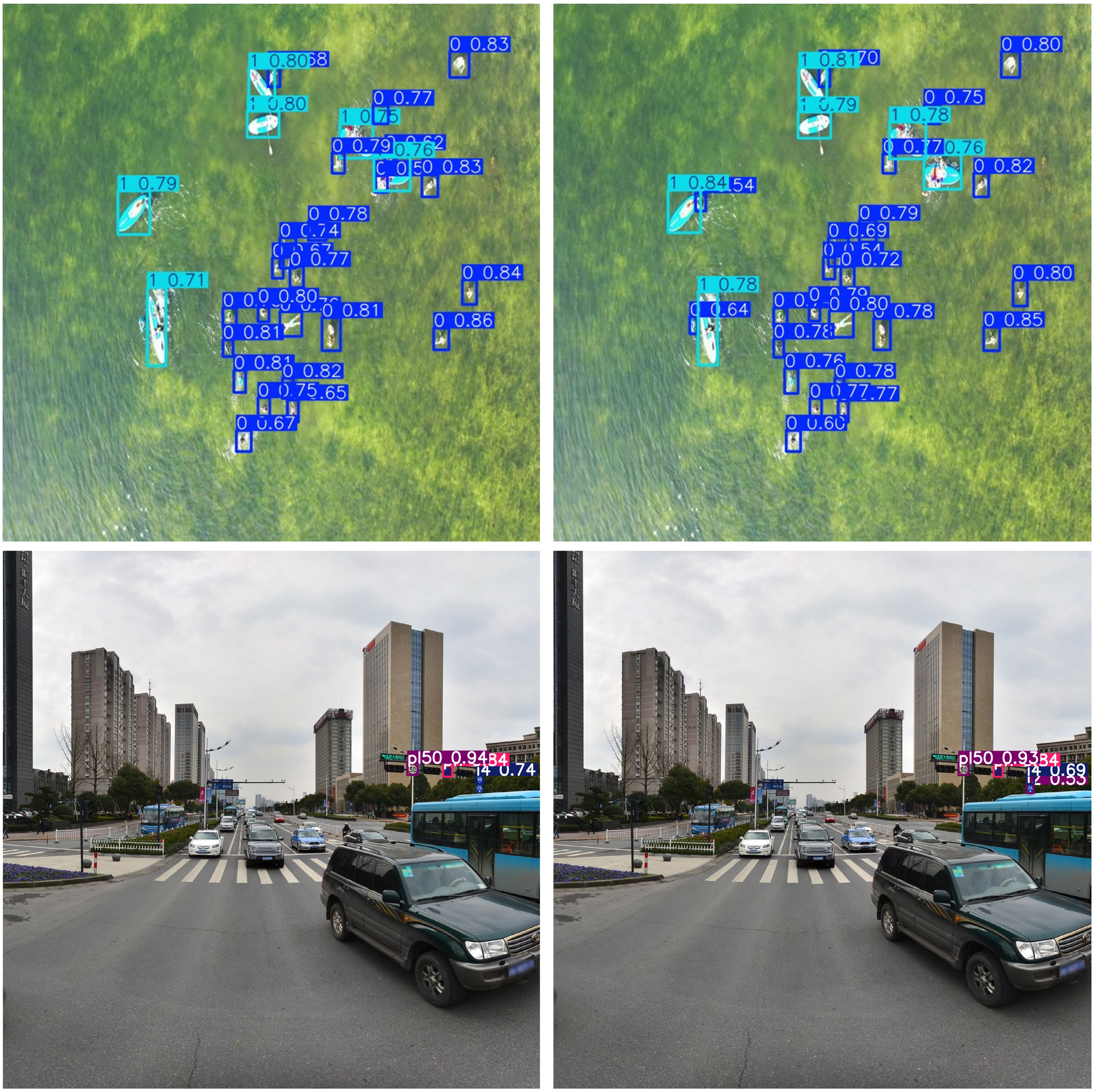
FIGURE 6. Object detection results of three different models on VisDrone2019 Dataset. (a) Results of YOLOv5s (b) Results of YOLOv8s (c) Results of DCE-YOLOv8.

3) INFERENCE TIME

Although the DCE-YOLOv8 maintains an acceptable inference time for real-time applications, it is still slower compared to YOLOv5 small and YOLOv8 small. This increase in inference time is a trade-off for the higher detection accuracy and reduced parameter count. For applications where speed is critical, this slight delay in processing might be a consideration. The proposed network utilizes depthwise operations on the most important blocks, such as DCE and SCDown. These operations require more memory

AFO

TT100K



(a) YOLOv8s (b) DCE-YOLOv8

FIGURE 7. Object detection results of YOLOv8s and DCE-YOLOv8 on AFO and TT100K Dataset.

than vanilla convolution at the inference stage. Convolution branching forces more memory to be used compared to normal operations that can be executed in a single process. Therefore, the modified proposed model employs fewer weights but requires more execution time than the original version.

4) GENERALIZATION TO DIFFERENT ENVIRONMENTS

Another potential weakness is the model's ability to generalize across different environments and conditions. While the DCE-YOLOv8 model has been trained on a diverse dataset, its performance might degrade in unseen or highly varied conditions. Figure 6 shows that in certain challenging environments, the detection accuracy can vary, indicating room for improvement in training methodologies and data augmentation approaches. Improving these aspects will allow the model to learn objects in low-light, foggy, and blurry environments.

In summary, while the DCE-YOLOv8 model demonstrates significant advancements in detection accuracy and efficiency, it is important to acknowledge and address these weaknesses. Several extreme case problems make it difficult for common detectors to recognize and localize objects precisely. These issues present opportunities for future work. Future research could focus on enhancing the model's robustness to false positives and negatives, improving small object detection, optimizing inference time, and ensuring better generalization across diverse environments.

VI. CONCLUSION

In this paper, we present an enhanced model by integrating Efficient Residual Bottleneck and Divided Context Extraction into YOLOv8. To mitigate computational complexity, we employ the Efficient Residual Bottleneck and eliminate the B5 block to improve the detection of small objects. Through Divided Context Extraction, we specifically target and extract features of small objects, leading to a notable enhancement in object detection rates based on these refined features.

The model is trained on the VisDrone dataset, yielding a mean Average Precision (mAP) of 32.7, which represents an approximate 43% improvement over the YOLOv8 small version. Additionally, the number of parameters is reduced by approximately 57%, culminating in a total of 4,822,382 parameters compared to the original YOLOv8.

The proposed detector offers several advantages, including improved detection of small objects, reduced computational load for real-time processing, increased accuracy, and flexibility for deployment on various hardware platforms. However, it also presents challenges such as the potential for overfitting, increased implementation complexity, compatibility issues with existing systems, and possible trade-offs between speed and accuracy. The proposed method demonstrates a higher mAP of 75.9 compared to YOLOv8's mAP.5:.95, with parameters totaling 4,855,654, which is 43% of that of YOLOv8 in the TT100K dataset. And in the AFO dataset, the model features only 2,080,421 parameters, approximately 18% of those in YOLOv8.

Across all three datasets, the model operates with real-time capability and exhibits superior object detection rates compared to all the networks being compared.

Future work will focus on further optimizing the model to balance speed and accuracy, ensuring it performs effectively in real-time applications. Additionally, we plan to explore advanced techniques for data augmentation and regularization to mitigate overfitting and enhance generalization across diverse datasets. Investigating the integration of the proposed model with existing object detection frameworks will also be a priority, as will assessing its performance in various real-world scenarios. Finally, we aim to extend the model's capabilities to handle dynamic environments and improve its robustness against occlusions and varying lighting conditions.

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